Functional Equations — problem 11.35

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A function f is defined on the set of rational numbers as follows :

$$f(0) = 0, \quad f(1) = 1, \quad f(x) = \begin{cases} f(2x)/4 & \text{for } 0 < x < \frac{1}{2}, \\ \frac{3}{4} + f(2x-1)/4 & \text{for } \frac{1}{2} \leq x < 1. \end{cases}$$

Let $a = 0.b_1b_2b_3\cdots$ be the binary representation of a. Find $f(a)$.